

AWS DOCUMENTATION:

AWS- (Amazon Web Services) is a cloud computing platform provided by Amazon.

USECASE-No need to purchase physical servers, Scalable, Pay-as-you-go, High availability, Global infrastructure, Automation, Security services.

AWS REGION- A **Region** is a geographical area containing AWS infrastructure.

USECASE- User location, Cost, Compliance, Service availability, Disaster recovery

AVAILABILITY ZONE- An **Availability Zone (AZ)** is an isolated location within an AWS Region.

Mumbai region – (AZ-A) (AZ-B) (AZ-C)

IAM – Identity and access Management

User, Group, Policy, Role

User – Particular user

Group- Collection of the users

Policy – Define permission's

Role – ec2 iam role s3

AUTHENTICATION – Who can access the account

AUTHORIZATION – What permission that user want

MFA – Multi Factor Authentication. Add second verification steps

LEAST PRIVILEGE - Give only the permissions required to perform the task.

EC2 (Elastic Compute Cloud) is an AWS service that provides **virtual servers in the cloud**.

Instead of buying and maintaining a physical server, you can create an EC2 instance in AWS and use it like a normal server.

AMI - Template used to launch an EC2 instance.

Instance Type-Determines CPU, memory, networking, etc.

Eg: t3.micro,t3.small,t3.medium

Key Pair-Used for secure authentication, commonly SSH access to Linux EC2.

ssh -i key.pem ubuntu@PUBLIC-IP

Security Group-Virtual firewall controlling network traffic.

SSH → 22

HTTP → 80

HTTPS → 443

Custom App → 8080

EBS

Persistent block storage attached to EC2.

EBS = Hard disk of your EC2 server

Public IP

Can provide internet-reachable connectivity when the network configuration allows it.

Private IP

Used for communication inside the AWS VPC/private network.

S3- simple storage service-Amazon S3 is an AWS object storage service used to store and retrieve data such as: Images, Videos, Documents, Backups Logs, Application files

Bucket - A **bucket** is a container where you store objects.

Object: An object is the actual file/data stored in an S3 bucket.

Object Key: A key is the unique identifier/name of an object inside a bucket.

S3 Region: The AWS Region where the S3 bucket's data is stored.

Types:

S3 Standard: For frequently accessed data.

S3 Intelligent-Tiering: Automatically moves objects between access tiers based on access patterns.

S3 Standard-IA: For infrequently accessed data that needs rapid retrieval.

S3 One Zone-IA: For infrequently accessed data stored in a single Availability Zone.

S3 Glacier Instant Retrieval: For archive data requiring millisecond retrieval.

S3 Glacier Flexible Retrieval: For archive data where retrieval can take minutes to hours.

S3 Glacier Deep Archive: For long-term archival with very low storage cost.

Versioning: Keeps multiple versions of an object.

Lifecycle Rules: Automatically transition or delete objects based on defined rules.

Bucket Policy: Controls access to an S3 bucket and its objects.

Block Public Access: Helps prevent unintended public access to S3 resources.

S3 Encryption: Protects S3 data at rest using encryption.

S3 Replication: Automatically copies objects from one S3 bucket to another bucket.

Static Website Hosting: Allows S3 to host static websites such as HTML, CSS, and JavaScript.

VPC - A VPC is a logically isolated virtual network in AWS where you can launch resources like EC2, RDS, and Load Balancers.

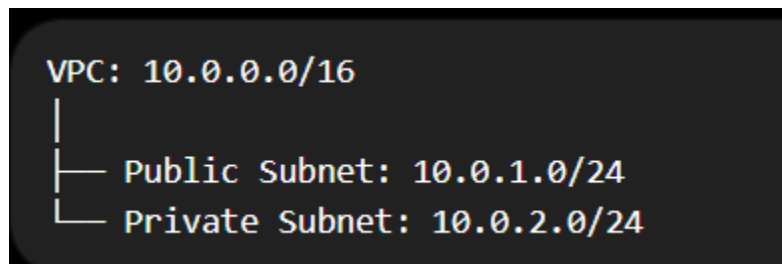
AWS

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└─ **VPC: 10.0.0.0/16**

CIDR - CIDR defines the IP address range of your VPC or subnet.

Subnet - A subnet is a smaller IP range inside a VPC.



Public subnet - A subnet is considered public when its route table has a route to an Internet Gateway.

Private Subnet - A private subnet doesn't have a direct route to an Internet Gateway.

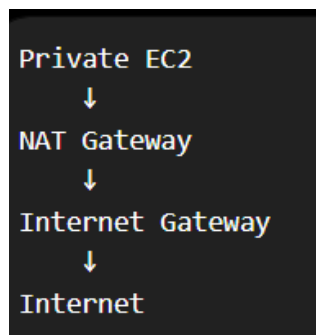
Internet Gateway - An Internet Gateway connects a VPC to the internet.

Route Table - A route table contains routing rules that determine where network traffic should go.

10.0.0.0/16 → traffic stays inside the VPC

0.0.0.0/0 → other traffic goes to the Internet Gateway

NAT Gateway - NAT Gateway allows resources in a private subnet to initiate outbound connections to the internet.



Security Group - A Security Group is a stateful virtual firewall attached to resources such as EC2.

SSH – 22

HTTPS – 443

HTTP -80

Network ACL - A Network ACL is a stateless firewall that operates at the subnet level.

